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**Paper title:** Finger joint angle and gesture estimation under natural conditions with a soft printed electrode array

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- Research field of the orientative paper.

Wearable biomedical sensing systems, specifically surface electromyography (sEMG) for neuromuscular monitoring and human-machine interface control.

### Summary of the paper

This paper presents a soft, screen-printed, flexible electrode array designed to record surface EMG signals from the forearm during natural, real-world finger movements. Unlike traditional gel or rigid electrodes, the printed electrode array conforms to the skin and remains comfortable during daily activity. The EMG signals collected from the array are processed using a deep learning model capable of estimating continuous finger joint angles and classifying discrete hand gestures. The study demonstrates that reliable finger movement decoding is possible outside controlled laboratory constraints, advancing wearable neuromuscular sensing toward practical everyday use.

### The results presented in the paper

- The soft printed electrode array maintained stable skin contact during natural movement (see *Image 1*), allowing reliable EMG signal acquisition without gels or rigid electrodes.
- A neural network model was trained to decode muscle activation patterns into:
  - Continuous finger joint angles (regression task)
  - Discrete gesture categories (classification task) (see *Image 2* showing the decoding architecture)
- Finger Joint Angle Estimation Results:
  - The system achieved  $R^2$  scores up to  $\sim 0.85$ , demonstrating strong correspondence between predicted and actual finger joint motion.
  - This confirms that the printed electrode array can capture subtle muscle activation differences needed for fine motor control.
- Gesture Classification Results:
  - The classifier reached gesture recognition accuracies up to  $\sim 0.87$  across subjects.
  - This indicates that distinct hand gestures generate identifiable EMG signal signatures, even during natural, non-constrained movement.
- Performance graphs (see *Image 3*) highlight:
  - Consistent decoding accuracy across repeated trials.
  - Inter-subject variability, suggesting that quick individualized calibration could enhance performance.
- Overall, the results show that soft, flexible EMG arrays can reliably support real-world hand movement estimation, moving EMG-based control systems closer to everyday prosthetic, rehabilitation, and human-machine interface use.

### Contribution (State of the Art Comparison)

- Unlike traditional EMG systems that use rigid or gel electrodes and require controlled hand movements, this paper demonstrates a soft, flexible printed electrode array that works during natural, everyday motion.
- It is one of the first works to simultaneously decode both continuous finger joint angles and discrete gestures from the same wearable EMG sensor array.
- This advances EMG-based control beyond laboratory settings, moving toward practical prosthetics, rehabilitation devices, and real-life human-machine interaction.

### Critical reasoning about the results explained in the paper

- The system's performance ( $R^2 \approx 0.85$  and accuracy  $\approx 0.87$ ) demonstrates strong decoding reliability under natural movement.
- However, results vary across users, meaning the model benefits from personal calibration for best performance.
- Long-term wearability and durability should be further tested in daily conditions.
- Overall, the paper provides a meaningful step toward practical, wearable EMG-based control systems.

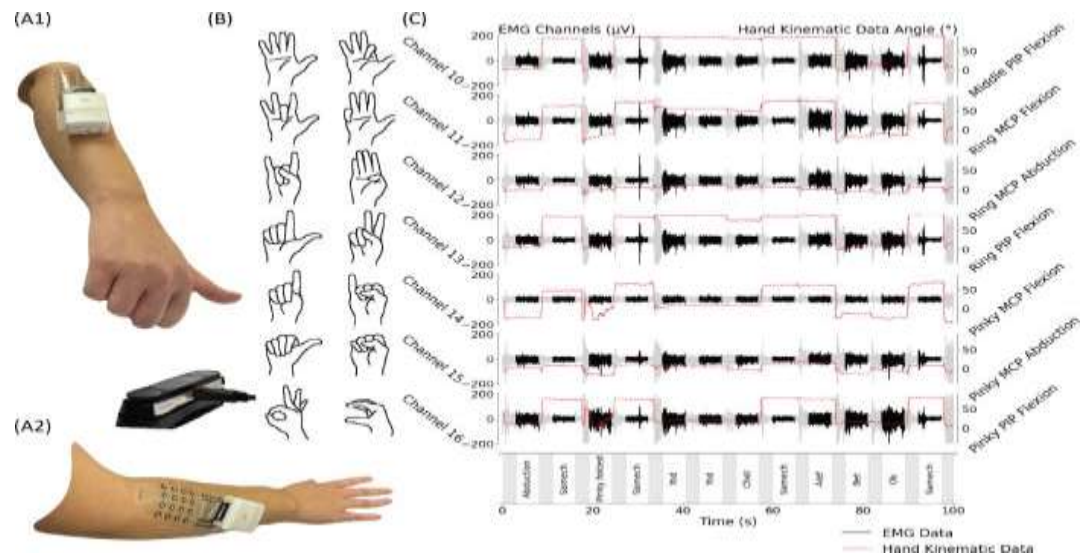


Image1

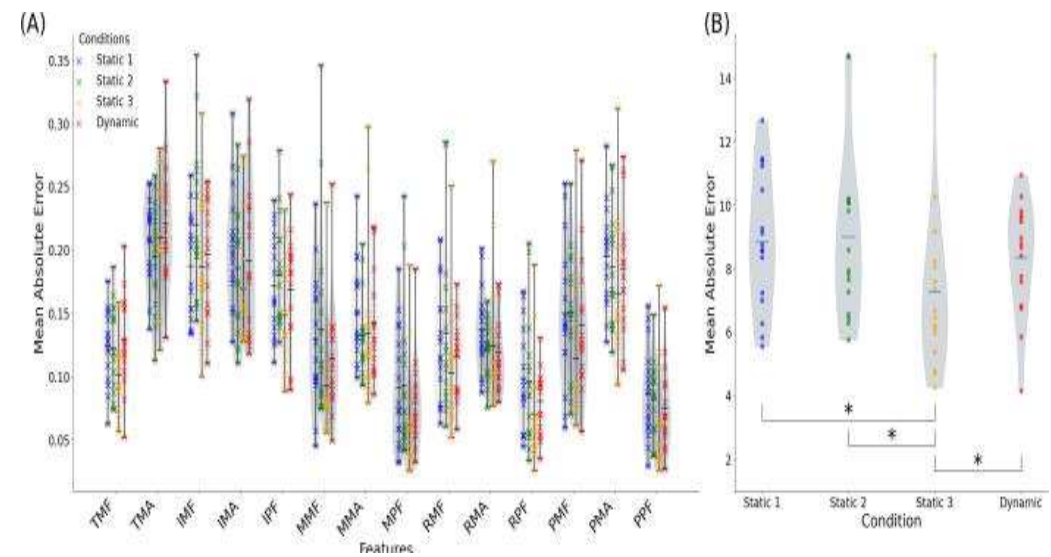


Image2

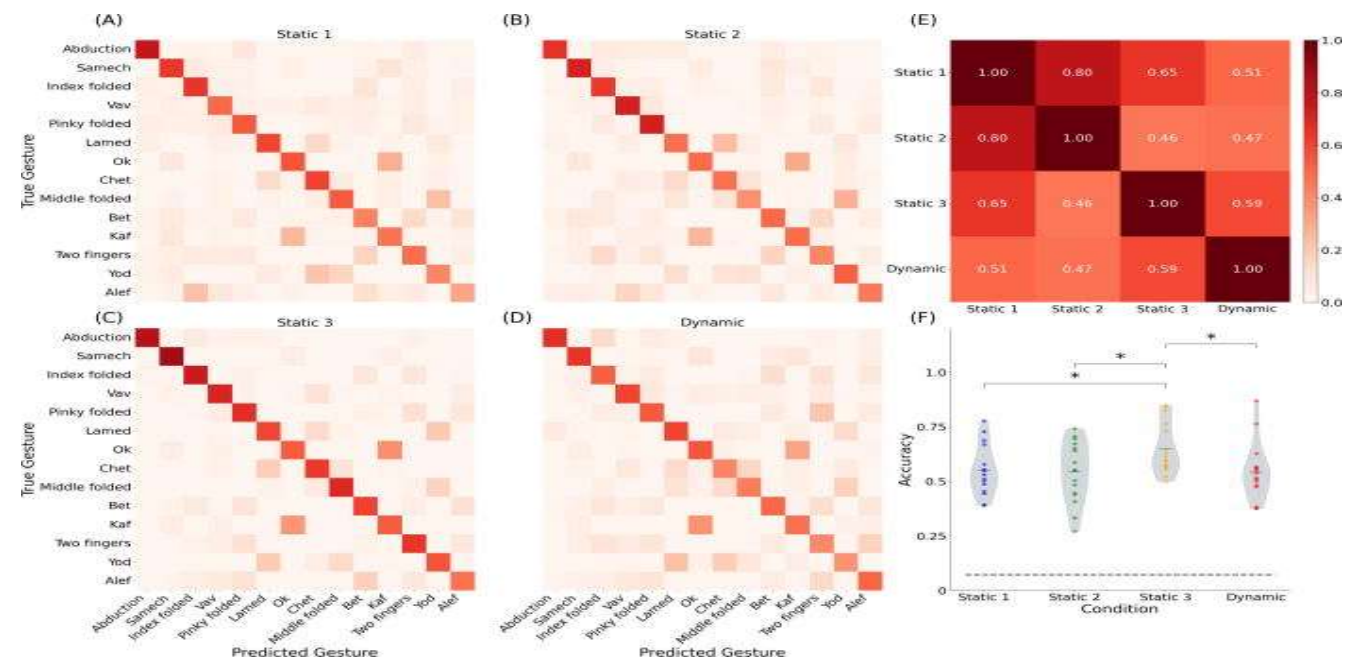


Image3